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Dear Dr. X,

Please consider our paper, entitled “Biological and climatic constraints drive reproductive synchrony” for publication as a X in *Nature Plants*. Here we answer a fundamental question in plant reproductive biology with relevance to climate change forecasting: what drives synchrony in seed production?

How plants time reproduction given unpredictable environmental conditions is a fundamental question in plant biology. This timing shapes not only individual reproductive success, but also the reproductive dynamics of whole populations and ecosystems. Across temperate and tropical forests, many species align their cycles so that most individuals reproduce together, driving seed crops that vary dramatically from year to year—a synchronous phenomenon called masting.

What sustains this synchrony is currently unknown. While climate is thought to synchronize reproduction, no climatic cue alone is sufficiently regular to explain the observed cycles[**LaMontagne2020**, **Journe2024**]. This suggests important missing mechanisms beyond climatic cues, which we identify and integrate here.

Building on decades of research on tree reproductive biology and long-term seed production data, we integrate a well-established reproductive constraint from fundamental plant and crop biology (rarely mentioned in masting studies): the temporal overlap between fruit maturation and floral bud initiation. We integrate this constraint alongside known climatic drivers of masting using an approach that links individual-tree reproduction to population-scale dynamics.

Our results show how climate and reproductive constraints together produce the classic ‘boom’ and ‘bust’ cycles of masting and determine synchrony. We show how cold summers act as a critical hinge, resetting all trees to a common reproductive state and driving long-term synchrony. In contrast to recent studies[**Bogdziewicz2020**, **Foest2024**] we find no evidence of shifts in masting, and show how reproductive constraints prevent runaway responses to warming. Our results highlight that any shifts in masting would come not from runaway seed production, but a decline in synchrony that requires first a decline in intermittent cold summers.

Our results advance fundamental plant biology by showing how the interplay between individual constraints and shared climatic cues provides a simple physiological mechanism for synchrony, with implications well beyond masting. Our model provides a new and much improved way to forecast how forests will regenerate as the climate warms. Beyond forests, we show how biological constraints can fundamentally limit the runaway amplification of climate effects at larger scales.

All authors contributed to this work and approve this version for submission. The manuscript is X words, with X references and X figures, and is not under consideration elsewhere. We hope you find it suitable for publication in *Nature Plants*, and look forward to hearing from you.

Sincerely,

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## References